

Unit 3 – Hands on Exercises for Variance, Sampling, Covariance and Correlation**1. Mean and Population Variance****Question**

The delivery times of five online orders are:

20, 25, 30, 35, 40 minutes

Calculate the mean and population variance.

Answer

Mean:

$$\text{Mean} = (20 + 25 + 30 + 35 + 40) / 5$$

$$\text{Mean} = 150 / 5 = 30$$

Mean = 30 minutes

Explanation

Population variance formula:

$$\sigma^2 = \Sigma(x - \mu)^2 / N$$

X	X – Mean	(X – Mean) ²
20	-10	100
25	-5	25
30	0	0
35	5	25
40	10	100

Sum of squared deviations:

$$100 + 25 + 0 + 25 + 100 = 250$$

Therefore,

$$\text{Variance} = 250 / 5 = 50$$

Result

Mean = 30 minutes

Population Variance = 50

2. Sampling

Question

A university has 5,000 students. A random sample of 100 students is selected to estimate the average daily study time. Explain why sampling is used instead of surveying all 5,000 students.

Answer

Sampling is used because surveying all 5,000 students would require more time, cost, and effort.

A random sample of 100 students can provide a reasonable estimate of the average study time of the entire university population.

Explanation

The university has:

- Population = 5,000 students
- Sample = 100 students

Sampling is preferred because:

- It saves time.
- It reduces cost.
- It requires less effort.
- A properly selected random sample can represent the population.
- Data collection and analysis become easier.

Result

A sample of 100 students is used because it provides an efficient way to estimate the average study time of 5,000 students while saving time and resources.

3. Sample Proportion

Question

From a sample of 200 customers, 150 customers are satisfied with a product. Calculate the sample proportion of satisfied customers and express it as a percentage.

Answer

Formula:

Sample Proportion = Number of satisfied customers / Total customers

$$= 150 / 200$$

$$= 0.75$$

Percentage:

$$0.75 \times 100 = 75\%$$

Explanation

Out of 200 customers, 150 are satisfied.

So, the proportion of satisfied customers is 0.75, which means 75% of the customers are satisfied.

Result

Sample Proportion = 0.75

Percentage of satisfied customers = 75%

4. Covariance

Question

Calculate the covariance between the following variables:

X	Y
2	5
4	7
6	9
8	11

Determine whether the relationship is positive or negative.

Answer

First calculate the means:

$$\bar{X} = (2 + 4 + 6 + 8) / 4 = 5$$

$$\bar{Y} = (5 + 7 + 9 + 11) / 4 = 8$$

Now calculate:

X	Y	$X - 5$	$Y - 8$	$(X - 5)(Y - 8)$
2	5	-3	-3	9
4	7	-1	-1	1
6	9	1	1	1
8	11	3	3	9

$$\Sigma(X - \bar{X})(Y - \bar{Y}) = 20$$

Population covariance:

$$\text{Cov}(X, Y) = 20 / 4 = 5$$

Explanation

The covariance is positive.

Positive covariance means that when X increases, Y also increases.

Here:

• X: $2 \rightarrow 4 \rightarrow 6 \rightarrow 8$

• Y: $5 \rightarrow 7 \rightarrow 9 \rightarrow 11$

Both variables increase together.

Result

Covariance = 5 Relationship = Positive

5. Exercise Hours and Calories Burned

Question

The following data represents hours of exercise (X) and calories burned (Y). Calculate the covariance and interpret the result.

X	Y
1	200
2	300
3	400
4	500
5	600

Answer

Calculate the means:

$$\bar{X} = (1 + 2 + 3 + 4 + 5) / 5 = 3$$

$$\bar{Y} = (200 + 300 + 400 + 500 + 600) / 5 = 400$$

Now calculate:

X	Y	$X - 3$	$Y - 400$	Product
1	200	-2	-200	400
2	300	-1	-100	100
3	400	0	0	0
4	500	1	100	100
5	600	2	200	400

$$\Sigma(X - \bar{X})(Y - \bar{Y}) = 1000$$

Population covariance:

$$\text{Cov}(X, Y) = 1000 / 5 = 200$$

Explanation

The covariance is positive (200).

This means that exercise hours and calories burned increase together.

As the number of exercise hours increases, the calories burned also increase.

Result

Covariance = 200

Relationship = Positive

Interpretation: More exercise hours are associated with more calories burned.

6. Pearson's Correlation Coefficient**Question**

Calculate Pearson's correlation coefficient for the following dataset and interpret the strength and direction of the relationship.

Study Hours (X)	Marks (Y)
1	45
2	50
3	55
4	65
5	70

Answer

Pearson correlation coefficient formula:

$$r = [n\sum XY - (\sum X)(\sum Y)] / \sqrt{[n\sum X^2 - (\sum X)^2][n\sum Y^2 - (\sum Y)^2]}$$

Values:

$$n = 5$$

$$\sum X = 15$$

$$\sum Y = 285$$

$$\sum X^2 = 55$$

$$\sum Y^2 = 16575$$

$$\sum XY = 925$$

Substituting:

$$r = [5(925) - (15)(285)] / \sqrt{[5(55) - 15^2][5(16575) - 285^2]}$$

$$r \approx 0.991$$

Explanation

The value of $r = 0.991$.

Pearson correlation ranges from -1 to +1.

- +1 → Perfect positive correlation
- 0 → No linear correlation
- -1 → Perfect negative correlation

Since 0.991 is very close to +1, there is a very strong positive correlation.

This means that as study hours increase, marks also tend to increase.

Result

Pearson's Correlation Coefficient = 0.991

Strength = Very Strong

Direction = Positive

Interpretation: Students who study more hours tend to obtain higher marks.